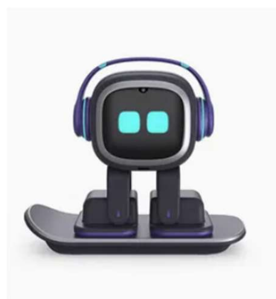
Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

CCE 322 :: Computer Peripheral and Interfacing Sessional



Project Title: EMO Robot

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Table of Contents

Emo Robot

1. Introduction

The Emo robot is an emotion detector robot that can detect people's feelings. We used a raspberry Pi 3 to create the project model. In this project we used a camera to capture human facial expressions and detect emotions. After detecting the emotion, the robot responds in many ways like it changes LED colors on screen, shows emotion related text or emoji on an OLED display, moves hands using servo motors and produce sound through a mini speaker.

The main goal of this project is to build a smart robot that can interact with humans in a more natural and expressive way.

2. Objectives

- To detect human emotions using a camera
- To show different LED colors for different emotions
- To display detected emotions on an OLED display
- To control the robot's hands using servo motors
- To produce sound on emotions

3. Problem Statement

People's feelings are a part of how we talk to each other. We can usually tell how someone is feeling by looking at their face. Machine and robots cannot do this.

In this project we aim to solve this problem by designing a robot that can recognize emotions and respond visually, physically and through sound.

4. Scope

The Emo robot project can be useful in many areas, including -

- Science fairs
- Exhibition
- Teaching in classroom
- Human-machine interaction research
- Robotics
- Research on how people and machines talk to each other.
- In old-age homes to be a companion if it can talk.

5. Methodology

We made the Emo robot by combining a few techniques and components. First the camera takes a picture of the person's face in real-time event. Then the Raspberry Pi 3 detects the picture of that person and detect the emotion that the person is feeling at that time. After that the robot does a thing based on how the person is feeling.

5.1 System Architecture

1. The camera takes a picture of the persons face
2. The Raspberry Pi 3 is the brain of the Emo robot
3. The RGB LED shows how the person is feeling using colors
4. The OLED Display Displays text or emoji (person's feeling)
5. The Servo Motors makes the hand movement of robot
6. The Mini Speaker makes sound
7. The Power Supply give the required power

Workflow:

Face Capture → Emotion Detection → Processing → Output Activation → Robot Response

5.2 Technology Stack

The parts we used to make the Emo robot are:

Hardware

- Raspberry Pi 3
- Pi Camera 5MP
- SG90 Servo Motors
- RGB LED
- 0.96" OLED Display
- Mini Speaker
- Breadboard
- Jumper Wires
- Resistors

Software / Libraries

- Python 3
- OpenCV
- DeepFace
- gpiozero
- luma.oled
- pygame

Programming Tool

VS Code for write the code

6. Costing Estimation:

Component	Price (৳)
Raspberry Pi 400 4GB RAM	11,300
Transcend 16GB UHS-I U1 MicroSD Memory Card	650
Raspberry Pi 4 Power Supply 5V 3A USB Type C	790
Servo Motor Micro SG90 180 Degree Rotation (2 pcs)	130*2
RGB Full Color LED SMD Module	55
220 Ohm 1/4W Resistors	10
0.96" Inch I2C OLED Display White	350
Breadboard Full Size Bare 830 Tie Points	145
Male to Female Jumper Wires 40 Pin 30cm	150
Raspberry Pi Camera 5MP	740
PCB Mount Mini Speaker 8 Ohm 0.2W	120
Others	500-1000
Total Cost	15,000-15,500/-

Components from: **RoboticsBd** (https://store.roboticsbd.com/)

7. **Gantt Chart**

Task	Month 1	Month 2	Month 3	Month 4
Structural Design	✓			
Circuit Assembly	✓	✓		
Sensor & Display Integration		✓	✓	
Emotion Coding			✓	✓
Final Assembly & Testing				✓

8. Visual Models

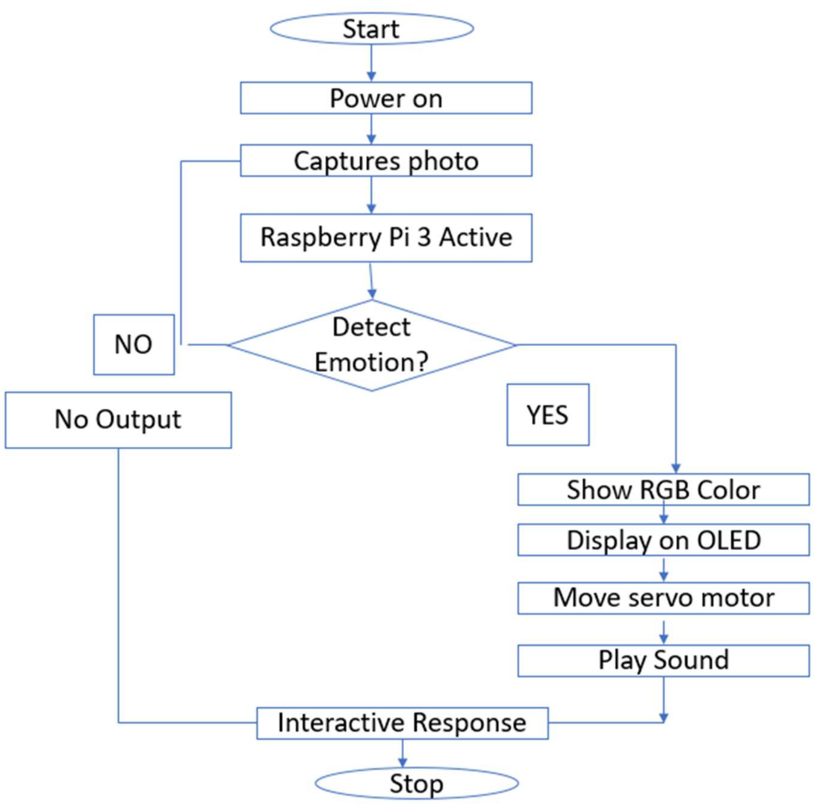

Fig: Flow Chart of Emo Robot

Block Diagram

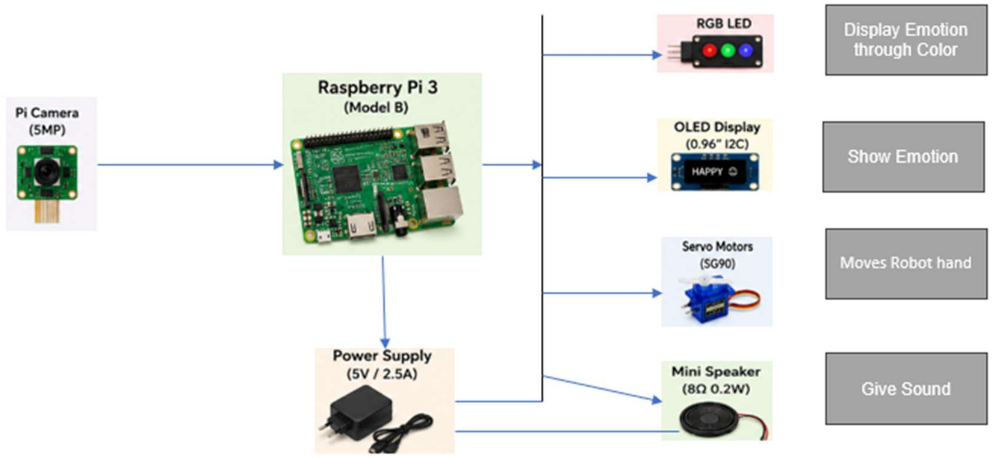

Fig: Block Diagram of Emo Robot

9. Future Plans

We can integrate the following features in future -

- Add more feelings like fear, tension, surprise, annoyance etc.
- Teach the robot to recognize gestures
- Make the robot respond to people with voice integration.
- Add mobile app control via wireless network.
- Improve accuracy for emotion detection.

10. Advantages

- Interactive robotic behavior
- Easy to understand
- Adjust AI and robotics
- Real time emotion detection

11. Limitations

The Emo robot is not perfect -

- It can't show how people are feeling if the light integration is bad.
- If the camera is not good the Emo robot does not work well.
- The Emo Robot can only tell feelings.
- The Emo Robot needs a good brain to work.

12. Conclusion

The Emo Robot project shows that a robot can tell how people are feeling and respond in a way that makes sense. With some problems the Emo Robot works well and can talk to people in a more natural way. It is an example of how robots can be used to help people.

13. References

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